Context-Aware IoT-Based Field Monitoring Module for Sri Lankan Paddy Cultivation Decision Support System

Project ID: R26-IT-066

Project Proposal Report

Bandara R.M. Asel Sahanjith– IT21811104

Individual component: **Intelligent Crop Recommendation System Integrating IoT, Climate Data, and Market Price Forecasts for Economic Viability in Sri Lankan Paddy Cultivation**

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Bandara R.M.A.S	IT21811104	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms.Chathurangika Kahandawarachchi	27/03/2026	
Ms. Buddhima Attanayake	27/03/2026	

ABSTRACT

Sri Lankan paddy cultivation plays a crucial role in national food security and rural livelihoods. However, farmers often face multiple challenges, including unpredictable weather conditions, inefficient resource management, crop diseases, and unstable market prices. Many farmers rely on traditional knowledge and static agricultural guidelines when deciding which crops to cultivate. These decisions rarely account for real-time field conditions or future market demand, leading to reduced productivity, inefficient use of water and fertilizers, and financial uncertainty. Existing crop recommendation systems exacerbate this issue by focusing primarily on soil and climate data while ignoring critical economic factors such as market price fluctuations and demand trends. Consequently, even crops that are agronomically suitable may not yield sufficient economic returns. Moreover, current systems are not well-adapted to local Sri Lankan conditions, such as monsoon variability, regional soil diversity, and paddy-specific cultivation practices. The absence of an integrated platform connecting IoT-based field monitoring, climate forecasts, and market intelligence creates fragmented decision-making and limits farmers' ability to make informed choices.

To address these challenges, this research proposes an intelligent decision support system for Sri Lankan paddy cultivation that integrates IoT sensing, climate data, and market price forecasting. The system collects real-time soil and environmental data using IoT sensors and combines them with climate information and historical market datasets. Advanced machine learning models analyse these inputs to recommend crops that are both environmentally suitable and economically profitable. By integrating field monitoring, climate prediction, and market intelligence into a single platform, the system enables optimized irrigation and fertilization schedules, reduces resource waste, mitigates financial risks, and supports dynamic, data-driven decision-making.

Solving this problem is crucial because it bridges the gap between agricultural productivity and economic viability. A system that provides reliable, real-time, and economically-informed crop recommendations can improve farmer livelihoods, stabilize rural incomes, and promote sustainable agricultural practices. Furthermore, at the commercial and national level, it contributes to more resilient supply chains, better resource management, and informed agricultural planning, ensuring that Sri Lankan paddy cultivation remains productive, profitable, and sustainable in the face of climatic and market uncertainties.

ACKNOWLEDGEMENT

We would like to take this opportunity to express our heart-felt thanks to all individuals and organizations who helped us in completing this research project.

First and foremost, we would like to express our highest gratitude to our project supervisor for their guidance, suggestions, and encouragement in completing this research project. We believe that their suggestions were very valuable in improving the quality of this research project.

We would also like to take this opportunity to express our thanks to our co-supervisor and all academic staff in the Information Technology department for providing us with all necessary knowledge required to successfully complete this research project.

Our heart-felt thanks go to all agricultural experts, farmers, and relevant authorities who shared their valuable knowledge regarding paddy cultivation in Sri Lanka. This information was very valuable in understanding real-world agricultural problems in relation to this research project.

Furthermore, we would also like to express our thanks to all relevant organizations that provide valuable information required to successfully complete this research project.

Lastly, we would also like to express our heart-felt thanks to all our families and friends for their encouragement in successfully completing this research project.

Their support was very valuable in helping us successfully complete this research project.

TABLE OF CONTENT (Auto Generated)

TABLE OF CONTENTS

TABLE OF CONTENTS 5
Table 1.1: Summery, Advantage and limitations of Literature Review 6
Table 1.2: Specific objectives and measurable outcome 6
Table 1.3: Software solution tools and Technologies 6
Figure 1.1: System Architecture of the Proposed Solution 7
Figure 1.2: Sensor Deployment Diagram of Paddy Field 7
Figure 1.3: Data Flow Diagram of Context Aware Interpretation Engine 7
Figure 1.4: Grant Chart 7
Figure 1.5: Work Breakdown Structure 7
Figure 1.6: Overall system diagram 7
1.1 Background and Context 7
1.2 Literature Review 8
1.3 Research Problem 9
1.4 Why we should solve the problem 9
1.5 Research gap 10
2.2 Specific Objectives 11
4.2 Individual Component Architecture 2

LIST OF TABLES

Table 1.1: Summery, Advantage and limitations of Literature Review

Table 1.2: Specific objectives and measurable outcome

Table 1.3: Software solution tools and Technologies

LIST OF ABBREVIATIONS

DSS - Decision Support System
IoT - Internet of Things
ML - Machine Learning
RF - Random Forest
XGB - XGBoost - Extreme Gradient Boosting
IRRI - International Rice Research Institute
SES - Standard Evaluation System
DOA - Department of Agriculture Sri Lanka
DCS - Department of Census and Statistics
API - Application Programming Interface
REST - Representational State Transfer
SDG - Sustainable Development Goal
WBS - Work Breakdown Structure
SLIIT - Sri Lanka Institute of Information Technology

LIST OF FIGURES

Figure 1.1: System Architecture of the Proposed Solution

Figure 1.2: Sensor Deployment Diagram of Paddy Field

Figure 1.3: Data Flow Diagram of Context Aware Interpretation Engine

Figure 1.4: Grant Chart

Figure 1.5: Work Breakdown Structure

Figure 1.6: Overall system diagram

1. INTRODUCTION

1.1 Background and Context

Agriculture plays a significant role in the Sri Lankan economy, and paddy farming is one of the key areas of agriculture. A large number of Sri Lankan farmers rely on rice farming as their principal source of income. However, paddy farming is faced with a number of problems, and the key problems faced by paddy farmers are unpredictable weather conditions, inefficient use of irrigation and fertilizers, diseases, and changes in the prices of crops. These factors often lead to a reduction in productivity, financial losses, and uncertainties in agricultural activities.

A key challenge faced by farmers today is the lack of data-driven decision-making. Most farmers often rely on their traditional knowledge and experiences to select crops for farming. However, they rarely consider the current environmental and financial conditions. This often results in financial losses, as the crops they cultivate may not yield the desired results, especially when the environmental conditions are unfavourable or the prices of crops vary significantly. In Sri Lanka, the agricultural sector contributes 7 to 8% to the country's GDP, and the agricultural sector provides employment to 25% of the population.

There are a number of technologies available to solve the problems faced by farmers. Smart farming using IoT technology can be used for the efficient use of irrigation systems, and machine learning algorithms have already been developed for crop yield prediction and disease detection using images. Additionally, agricultural guidelines provided by government institutions offer recommended fertilizer schedules and cultivation practices.

Despite these technological achievements, most existing smart agriculture systems are still isolated and only cover limited aspects of agriculture. For instance, existing crop recommendation systems are mostly based on soil conditions and climatic conditions, without considering economic factors such as market demand and price changes. Similarly, existing IoT-based smart agriculture monitoring systems are mostly based on providing raw data from the field without any intelligent decision-support tools. Therefore, farmers are still not able to make informed decisions that take into consideration both environmental and economic factors.

Hence, to overcome these limitations, this research aims to develop an integrated smart agriculture decision support system based on IoT, climate analysis, and market price prediction. The proposed system will be able to provide the best crop recommendations to farmers by analysing real-time soil and environmental conditions along with historical and predicted trends related to market conditions. Machine learning algorithms will be used to analyse these datasets and provide intelligent crop recommendations. Therefore, the proposed system will be able to integrate all these sources of information to provide a unified platform for farmers to make informed decisions.

This research will also be aligned with the United Nations' Sustainable Development Goals (SDGs), which include SDG 2: Zero Hunger, aiming to end hunger and ensure access to food by promoting sustainable agriculture. Similarly, this research will be aligned with SDG 12: Responsible Consumption and Production, which encourages the use of resources efficiently. Therefore, this research will be aligned with these two SDGs by improving agricultural production and reducing economic risks for farmers.

Moreover, this research will be aligned with the Software Systems and Technologies (SST) research cluster because this research involves the development of a smart agriculture decision support system based on advanced software systems and intelligent technologies.

1.2 Literature Review

A systematic review of 9 scholarly sources was conducted to evaluate the current state of research in intelligent crop recommendation systems, with a focus on the integration of soil parameters, climate conditions analysis. The review examined how machine learning algorithms and IoT-based data collection techniques are used to generate accurate and economically viable crop recommendations. Particular attention was given to studies incorporating real-time sensor data such as soil pH, moisture, and nutrient levels (NPK), alongside environmental factors like temperature and rainfall. Additionally, the review analysed approaches that integrate market trends to enhance profitability and decision-making for farmers. The findings highlight the growing importance of data-driven agricultural systems in improving crop selection, optimizing resource utilization, and increasing farmer income through informed, context-aware recommendations.

Study Areas	**Technologies**	**Features**	**Advantage**	**Limitations**
Traditional farming method	Farmer experience, static guidelines	Manual crop selection based on past knowledge	Simple, Low cost	Low accuracy, No data driven decisions
ML based crop recommendation	ML(Random forest)	Predict crops using soil nutrients(NPK), ph, temperature	High prediction accuracy, Better planning	Uses static data, Lacks real-time updates
Iot based monitoring system	Iot sensors	Real-time monitoring of soil moisture, Temperature, humidity	Real-time data collection, Improved resource management	Focus mainly on monitoring, limited decision support
Precision agriculture system	Iot + data analytics	Integration of sensor data with analytics for resource optimization	Efficient water and fertilizer usage	Limited focus on profitability
Big data agriculture systems	Big data	Combine multiple data sources (weather, sensors)	Comprehensive analysis, scalable systems	Complex implementation, high infrastructure cost

Deep learning approaches	Neural networks, AI	Advanced analysis for yield prediction, Disease detection	High accuracy, Handle complex datasets	Requires large datasets, high computational cost
Existing crop recommendation system	ML + Environmental Data	Suggest crops based on soil and climate conditions	Environmentally suitable recommendations	Ignore market prices, not economically optimized
Market Analysis Systems	Market databases, Price trend analysis	Analyse supply demand and price trend	Helps in profitability decisions	Not integrated with environmental data
Proposed integrated system	Iot + ML + Market Analytics	Combines real-time data, Prediction models	Accurate, Real-time economically viable recommendations	Requires integration and system complexity

1.3 Research Problem

In Sri Lankan agriculture, crop selection decisions are often based on traditional knowledge and generalized guidelines rather than data-driven insights. Existing crop recommendation systems primarily rely on static soil properties and historical climate data, without considering real-time field conditions or economic factors. This leads to suboptimal crop choices, inefficient resource utilization, and unstable farmer income.

A major limitation in current approaches is the lack of integration between **IoT-based field data, dynamic climate conditions, and market price fluctuations**. Most systems recommend crops based only on agronomic suitability (e.g., soil type, rainfall), ignoring whether the selected crop will be economically profitable at harvest time. As a result, farmers may grow crops that are suitable for the land but yield low financial returns due to oversupply or unfavourable market conditions.

Furthermore, existing models often fail to adapt to **localized Sri Lankan conditions**, such as monsoon variability, regional soil diversity, and paddy-specific cultivation practices. The absence of real-time data integration and predictive market intelligence creates a gap between agricultural productivity and financial viability.

Therefore, there is a need for an **intelligent, integrated crop recommendation system** that combines real-time IoT sensor data, climate forecasts. Such a system can support farmers in making informed decisions by recommending crops that are not only suitable for current field conditions but also optimized for maximum economic return, reducing risk and improving sustainability.

1.4 Why we should solve the problem

Climate change makes traditional farming knowledge unreliable

Farmers historically relied on patterns passed down through generations — seasonal cues, rainfall timing, temperature norms. Climate change has disrupted these patterns profoundly. Rainfall is less predictable, temperatures are shifting, and drought or flood events are more frequent and intense. A farmer who planted rice every March for 40 years can no longer trust that same calendar. An AI-integrated system that reads real-time weather forecasts and climate trends fills the knowledge vacuum that climate change has created.

The scale of loss is enormous and avoidable

Globally, crop failure and mismatch between soil/climate conditions and planted crops waste billions in agricultural inputs every year — seeds, fertilizer, water, and labour. In developing economies, this loss is felt most acutely because farmers cannot afford to experiment. The tragedy is that a large portion of this loss is preventable with better information. Your system targets exactly this gap: not more resources for farmers, but smarter use of the resources they already have.

Human survival and food security

Agriculture feeds humanity. When farmers make wrong crop decisions, entire families and communities go hungry. Across South and Southeast Asia — including Sri Lanka — millions of smallholder farmers operate on razor-thin margins with no buffer against a failed season. A single bad harvest can push a family from subsistence into debt or food insecurity. The problem is not laziness or ignorance; it is a lack of timely, reliable, location-specific guidance. Solving this directly protects human welfare at its most basic level.

1.5 Research gap

G1 — Absence of Market-Aware Crop Recommendation for Sri Lankan Paddy No existing crop recommendation system in Sri Lanka integrates market supply-demand forecasting with agronomic field assessment. Tools such as eKeth, GeoGoviya, and Krushi Advisor provide seasonal crop calendars and generic agronomic advisories based on static data, but none incorporate future market price prediction or supply-demand pressure into the crop selection decision. Farmers therefore make planting decisions with no visibility into whether their chosen crop will be profitable at harvest time, directly contributing to income instability among smallholder paddy farmers.

G2 — Lack of Real-Time IoT Integration in Sri Lankan Crop Advisory Systems Existing Sri Lankan crop advisory platforms rely entirely on static historical datasets, printed DOA guidelines, or manually entered field parameters. No current system ingests live IoT sensor readings of soil pH, moisture, NPK, temperature, and humidity to dynamically assess field suitability at the time of the recommendation. This means recommendations do not reflect actual current field conditions, reducing their accuracy and practical relevance for individual farmers whose field conditions vary significantly across plots and seasons.

G3 — Non-Localized ML Models with No Sri Lankan Paddy Specificity Reviewed ML-based crop recommendation and yield prediction models such as Abdurrahman & Miyim (2025) and Cui et al. (2023) are trained on datasets from Nigeria, India, or other Asian contexts and do not account for Sri Lankan-specific conditions including BG 300 and AT 362 variety characteristics, Maha and Yala

seasonal dynamics, local monsoon rainfall patterns, and Sri Lankan soil profiles. Applying these models directly to Sri Lankan paddy fields produces systematically inaccurate recommendations due to the mismatch between training data distribution and local agronomic reality.

2. OBJECTIVE

To develop a market-aware, IoT-integrated intelligent crop recommendation system for Sri Lankan paddy farmers that predicts the most economically viable crop to cultivate by fusing real-time IoT sensor data (soil pH, moisture, NPK, humidity) from the field monitoring module, local climate data (rainfall, temperature, monsoon patterns) from the Sri Lanka Meteorological Department, and market supply/demand forecasts from agrarian databases — achieving a crop recommendation accuracy of 85% or above and a market viability prediction R-squared of 0.80 or above and delivering a ranked three-scenario crop recommendation (high profit, balanced risk, agronomic safe choice) through a web-based dashboard to enable evidence-based planting decisions for Sri Lankan smallholder paddy farmers.

.

2.2 Specific Objectives

Gap	Specific Objective	Measurable Success Criteria
G2	**SO1** — To develop a machine learning based agronomic suitability classification model using Random Forest and XGBoost that predicts the most suitable paddy crop type and variety for a given field using real-time IoT sensor inputs (soil pH, moisture, NPK, temperature, humidity) received from the field monitoring module and local climate data (rainfall, temperature) from the Sri Lanka Meteorological Department API, trained specifically on Sri Lankan paddy variety data including BG 300 and AT 362	Classification accuracy ≥ 85% on held-out Sri Lankan DOA district field validation dataset for BG 300 and AT 362 varieties across Maha and Yala seasons; F1-score ≥ 0.83 across all suitability classes
G1, G3	**SO2** — To develop a market viability prediction regression model using	R-squared ≥ 0.80 on held-out Sri Lankan market price validation dataset spanning at least three

	XGBoost Regressor and LSTM that forecasts future paddy market price and supply-demand pressure index using historical market data from the Hector Kobbekaduwa Agrarian Research Institute and government agrarian databases, fine-tuned for Sri Lankan district-level paddy price seasonality patterns and Maha/Yala supply cycle characteristics	consecutive Maha/Yala seasons; market demand pressure index prediction statistically significant with $p < 0.05$ on season-price interaction term
G4	**SO3** — To integrate the agronomic suitability model and market viability prediction model into a three-scenario ranked recommendation engine that generates parallel crop recommendations (High Profit, Balanced Risk, Agronomic Safe Choice) with associated net revenue projections per acre using market price forecasts from SO2 and input cost references from the DOA pesticide and fertilizer cost reference table stored in the system database	Recommendation engine generates all three ranked scenario outputs within 3 seconds of receiving input data; net revenue projection validated against at least two historical Maha season market price records from the Hector Kobbekaduwa Agrarian Research Institute database; scenario ranking consistency ≥ 90% across repeated inference runs on the same input
G4	**SO4** — To develop a React.js web dashboard that displays real-time three-scenario ranked crop recommendations, net revenue projections per acre, field suitability status indicators, and market viability scores integrated with the group unified platform via REST API in both Sinhala and English, auto-refreshing on new IoT sensor data receipt or updated market price signals	Dashboard passes usability testing with ≥ 80% task completion rate among 10 Sri Lankan smallholder paddy farmer participants in Polonnaruwa or Kurunegala district; FastAPI REST API response time ≤ 3 seconds under standard network conditions; bilingual Sinhala/English toggle verified functional across all dashboard components

3. METHODOLOGY

This research adopts a Design Science Research (DSR) methodology, combining iterative machine learning model development, experimental validation, and system engineering. DSR is appropriate because the primary output is a working artifact — the multi-stage paddy crop type recommendation system — whose quality is evaluated against defined performance criteria. The approach proceeds across five phases, each mapped to a specific research objective.

3.1 Research Approach

The research follows the DSR cycle: (1) Problem identification and motivation through literature review and gap analysis identifying the absence of data-driven, field-specific paddy variety selection tools in Sri Lankan precision agriculture; (2) Definition of objectives for a three-stage suitability-based recommendation output; (3) Design and development across Phases 2–4 below; (4) Demonstration through a functional prototype recommendation component; and (5) Evaluation through model validation and agronomist usability testing. This cycle ensures that the artifact design remains grounded in the identified research gap and that evaluation criteria are pre-defined and measurable.

1) Data collection

First: real-time soil data will be received from the IoT sensor component of this research system, providing continuous field-level readings of soil moisture, soil temperature, nitrogen (N), phosphorus (P), potassium (K), pH. These sensor readings represent the live soil condition profile of the target paddy field and form the primary soil input layer of the recommendation engine.

Second: weather and climate data will be sourced from meteorological APIs — specifically Department of Meteorology Sri Lanka — providing historical and current seasonal records of daily temperature (minimum, maximum, mean), total rainfall (mm), relative humidity (%), sunshine hours per days. District-level seasonal climate records spanning the Maha and Yala cultivation seasons will be incorporated to capture seasonal variability in suitability scoring.

Paddy variety: A paddy variety reference database will be constructed by digitizing agronomic requirement profiles for primary Sri Lankan paddy varieties — including BG 352, BG 358, BG 360, BG 94-1, At 362, and BG 403 — from the Department of Agriculture Sri Lanka variety release documentation and IRRI rice knowledge bank, producing documented optimal ranges for every soil and climate parameter per variety. This database serves as the ground-truth reference for suitability scoring.

2) Suitability Scoring Model

A Weighted Suitability Index (WSI) will be computed for each candidate paddy variety against the field's current soil and climate profile. The WSI formula is

$$\mathbf{WSI = \Sigma\ (W_i \times S_i) / \Sigma\ W_i}$$

where W_i is the agronomic importance weight of parameter i and S_i is the individual suitability score (0–100) for that parameter, derived by comparing the observed sensor or climate value against the variety's documented optimal range in the reference database.

The success criterion for Phase 2 is that the WSI correctly classifies field-variety combinations into their agronomist-validated suitability category with an accuracy of **≥ 85%** on the held-out validation dataset.

3) **Hybrid Recommendation Engine**

   Varieties that violate minimum agronomic thresholds for critical parameters — such as soil pH falling below 4.5, seasonal rainfall below the varietal minimum, or temperature exceeding the critical upper threshold — are immediately eliminated as Not Suitable (N) before entering the ML ranking stage. This rule layer ensures agronomic correctness and prevents computationally plausible but agronomically infeasible recommendations.

   The remaining candidate varieties are ranked by a trained Random Forest and XGBoost ensemble classifier trained on historical agricultural field records matched with known suitability and yield outcome labels sourced from the Department of Census and Statistics district-level seasonal records and Department of Agriculture field trial data. Hyperparameter tuning will be performed via 5-fold cross-validation on the training set. Model performance will be reported on a held-out test set using Accuracy, Precision, Recall, and F1-Score. The success criterion is F1-Score ≥ 0.85.

   The top-ranked varieties are assigned to three suitability stages based on their computed WSI scores. Varieties scoring 80% and above are assigned to **Stage** 1 (S1 — Highly Suitable). Varieties scoring between 55% and 79% are assigned to Stage 2 (S2 — Moderately Suitable). Varieties scoring between 35% and 54% are assigned to Stage 3 (S3 — Marginally Suitable). Varieties scoring below 35% are classified as Not Suitable and excluded from the output.

4) **Recommendation output**

   These outputs generate a structured recommendation report for each query field. Each stage output will include the recommended paddy variety name, its WSI score, identified limiting factors, and targeted management suggestions addressing specific parameter deficiencies detected in the soil and climate data — for example, recommending lime application where pH is marginally low for an S2-classified variety.

Software Solution Tools and Technologies

Category	Tool	Justification

Field data collection (real-time)	ESP32 Microcontrollers and Sensors	ESP32 microcontrollers integrated with sensors (such as soil pH, NPK, temperature, and humidity sensors) are used to collect real-time field data directly from the agricultural environment.
Project management	GitHub	used for version control and collaboration, allowing developers to manage source code, track changes, and work together efficiently when building the crop recommendation system.
Main AI backend service	Python with FastAPI	used to build a high-performance backend API that processes sensor data and delivers real-time crop recommendations quickly and efficiently.
Real-time communication	NodeJs	used to handle server-side operations and real-time data communication, enabling smooth interaction between the frontend, database, and IoT devices.
Store all system data	Postgre SQL Databases	used to store and manage structured data such as soil parameters, weather data, and crop details, ensuring reliable and scalable data handling
Crop recommendation model	TensorFlow	used to develop and deploy machine learning models that analyze soil and climate data to predict the most suitable crops for cultivation.
Website hosting	Hostinger KVM01 VPS	used to host the entire system, including APIs, databases, and machine learning models, providing a stable and scalable environment for real-time crop recommendations.

Commercialization

The Intelligent Crop Recommendation System integrates IoT sensor data (soil pH, NPK, moisture), climate information, and real-time market prices to recommend the most suitable and profitable crops. The goal is to improve farmers' productivity while ensuring economic viability.

End Users

Primary Users

Small & medium-scale paddy farmers in Sri Lanka who make seasonal planting decisions and need guidance on what to grow for maximum profit

Agricultural cooperative societies that advise clusters of farmers across districts

Secondary Users

Agri-input suppliers (fertilizer, seed companies) who want to anticipate demand by region

Government agricultural extension officers who assist farmers in making crop decisions

Agri-finance institutions & microfinance banks that need crop viability data before issuing farm loans

Export/wholesale buyers & traders who want early signals on upcoming supply volumes

Free version - Component

Soil suitability check — farmer inputs basic field details (region, soil type) and gets a simple yes/no on whether paddy is suitable for their land

Top 2 crop suggestions — a simplified recommendation showing the top two crops suited for that season, based on historical averages

Limitations of free tier

No real-time IoT sensor integration

No demand forecasting or profit margin prediction

Paid version - Component

Supply & demand forecasting — ML model predicts future market prices and surplus/shortage periods so the farmer can time their harvest for peak prices

Top 5 ranked crop recommendations — ranked by both agronomic suitability and projected profitability, not just soil fit

Future Scope

- Expansion to Other Crops: Adapt the context-aware monitoring framework for other economically important crops in Sri Lanka including tea, rubber, coconut, and vegetables by developing crop-specific knowledge bases and growth stage models.
- AI-Driven Disease Prediction: Enhance the interpretation engine with machine learning models that predict disease outbreaks by analyzing correlations between environmental conditions, crop growth stages, and historical disease incidence patterns.
- Smart Advisory Expansion: Incorporate voice-based recommendations in local dialects and video-based advisory content to improve accessibility for farmers with limited literacy and enhance knowledge transfer effectiveness.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the Proposed Solution and its Components

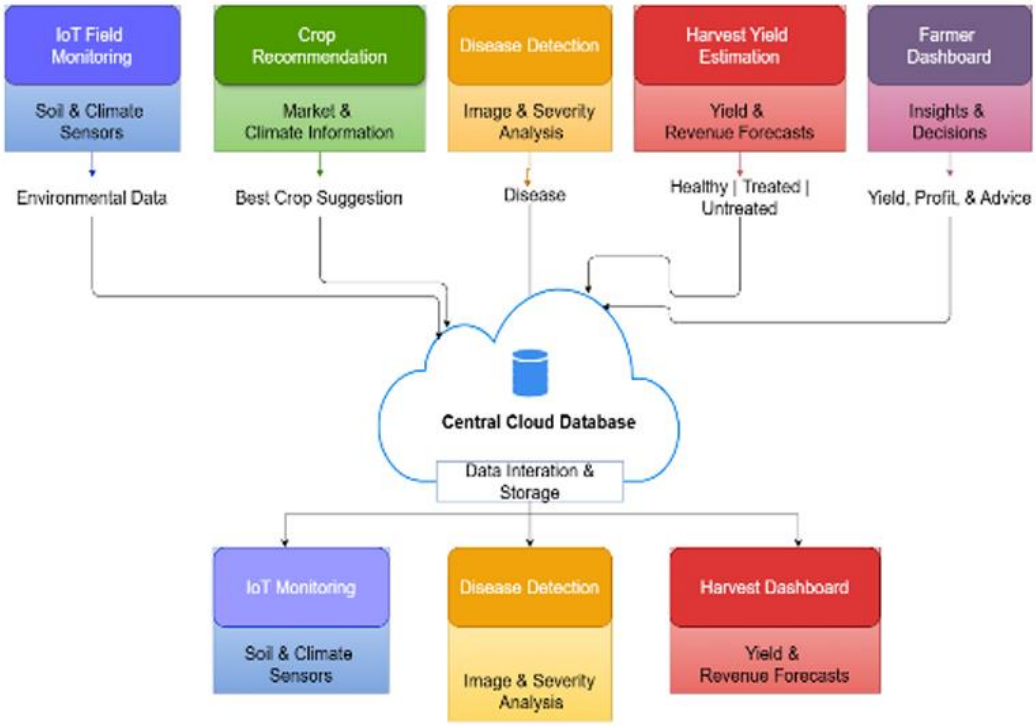
The proposed system is a four-module Smart Agriculture DSS for Sri Lankan Paddy Cultivation (Project R26-IT-066). Each module addresses a distinct problem domain and communicates with others via REST APIs through a shared cloud layer. The four modules and their relationships are:

• IT22178718 - Context-Aware IoT Field Monitoring: Continuously collects real time soil and environmental sensor data (moisture, pH, NPK, temperature, humidity, water level). Sends processed sensor readings to this component (IT22266446) as inputs to the base yield estimation model.

• IT21811104 - Intelligent Crop Recommendation: Analyses IoT and climate data against market supply/demand forecasts. Sends current paddy market price per kg to this component for revenue projection calculations within the scenario engine.

• IT22261632 - Disease Detection and Severity Analysis: Detects Rice Blast and Brown Spot from leaf images and environmental sensor data. Estimates disease severity percentage and growth stage at detection. Sends disease type, severity score, and growth stage to this component as inputs to the disease-yield loss regression models.

• IT22266446 - Disease-Aware Harvest Yield Estimation: Receives IoT data, disease severity scores, and market price from the three other modules. Applies the base yield model and disease-yield loss models to generate three-scenario yield and revenue forecasts with net treatment benefit calculation. Delivers outputs via REST API to the unified group dashboard.



4.2 Individual Component Architecture

The crop recommendation component consists of four internal processing layers, each directly aligned with a specific objective

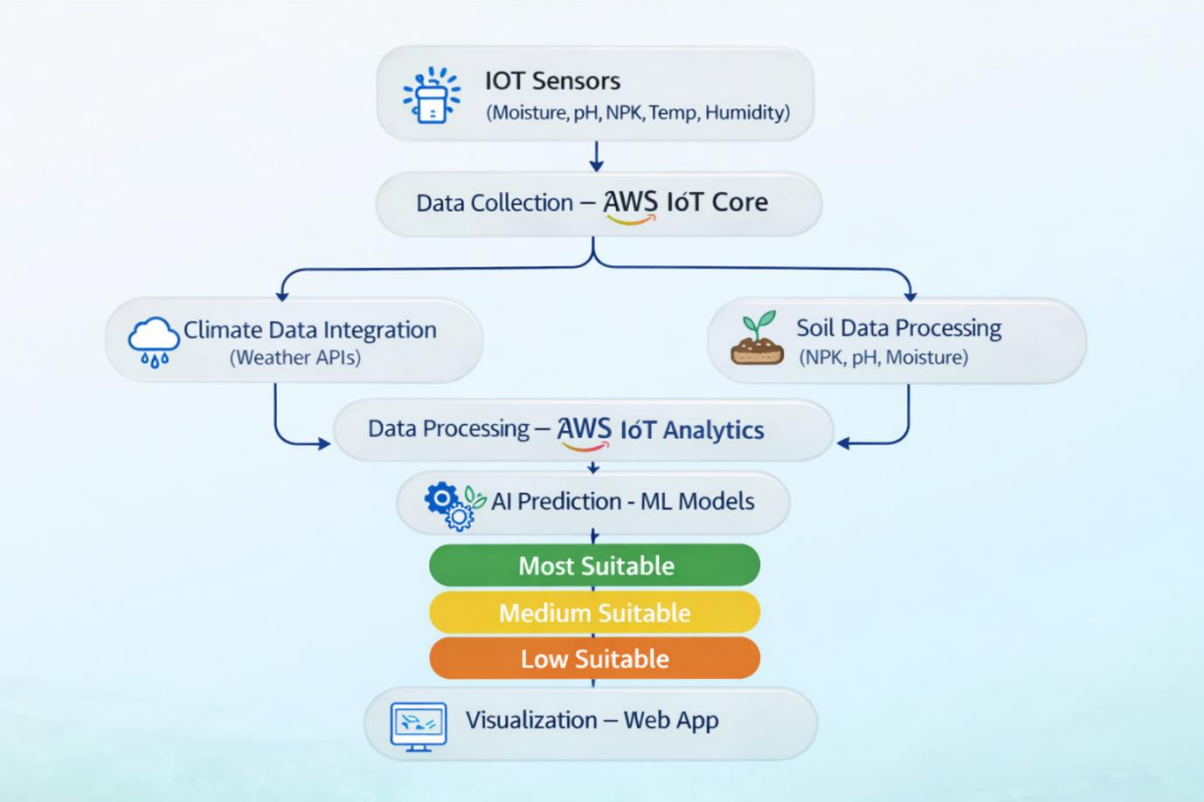
Data sources — Field IoT sensors stream soil parameters (moisture, pH, NPK, temperature, humidity) into AWS IoT Core, which authenticates devices and fans the data into two parallel tracks.

Parallel ingestion — One track pulls real-time and forecast weather from the Sri Lanka Meteorological Department API; the other extracts, cleans, and normalizes NPK, pH, and moisture into a structured soil profile. Both outputs converge in AWS IoT Analytics.

Analytics and feature engineering — The merged stream is cleaned (outlier detection, imputation, timestamp alignment, normalization) and enriched with composite features — soil fertility index and agro-climatic suitability score — producing a single structured input vector for the model.

Prediction — An ML model trained on Sri Lankan paddy data scores and ranks candidate crops into three tiers. Market price signals then re-rank results by profitability, so the final recommendation reflects both agronomic fit and financial return.

Farmer interface — A React.js dashboard presents color-coded recommendation cards with Sinhala and English explanations, a profit projection panel showing estimated revenue per acre, and auto-refreshes on any new sensor or price data.



5. USER REQUIREMENTS

Stakeholder Identification

1. **Paddy Farmers (Primary Users)**
   Farmers use the system to receive crop recommendations based on **IoT field data, climate conditions, and predicted market prices** to maximize profitability and reduce cultivation risk.

2. **Agricultural Officers / Agronomists**
   Experts who provide guidance and verify the system's recommendations using agricultural knowledge and field experience.

3. **Government Agricultural Organizations**
   Institutions such as the Department of Agriculture that provide **crop guidelines, climate data, and agricultural policies** used by the system.

4. **Market Data Providers**
   Organizations or research institutes that provide **historical and real-time crop price and demand data** used for economic analysis.

TAF-R26-IT-066-

6. System Administrators / Developers

Responsible for maintaining the system, updating models, managing datasets, and ensuring the recommendation engine operates correctly.

Functional Requirements

FR1. The system shall **collect IoT sensor data** such as soil moisture, temperature, humidity, and pH from the field.

FR2. The system shall **integrate climate and weather data** to analyse environmental conditions affecting crop growth.

FR3. The system shall **analyse sensor data and climate conditions** to determine crop suitability for the field.

FR4. The system shall **apply machine learning models** to predict crop profitability based on environmental and market data.

FR5. The system shall **generate crop recommendations** ranked according to expected economic return.

FR6. The system shall **display recommended crops and supporting insights** through a dashboard interface.

FR7. The system shall **allow farmers to field information** to personalize recommendations.

FR8. The system shall **update recommendations periodically** when new IoT, climate, or becomes available.

FR9. The system shall **suggestions** if the recommended crop is not suitable for the current field conditions.

Non-Functional Requirements

1. Security

- The system shall ensure **secure storage of farmer data and sensor information**.
- Access to system management functions shall be **restricted to authorized administrators**.

2. Performance

- The system shall **process IoT and climate data efficiently** to generate recommendations within a reasonable response time.
- The system shall support **real-time or near-real-time data updates**.

3. Usability

- The user interface shall be **simple and easy for farmers to understand**.
- The dashboard shall display **clear recommendations and explanations**.

4. Reliability

- The system shall provide **accurate and consistent crop recommendations** based on validated data sources.
- The system shall continue operating even if some sensor data is temporarily unavailable.

5. Scalability

- The system shall support **multiple farms and users simultaneously**.
- The architecture shall allow integration of **additional datasets or future modules**.

6. Maintainability

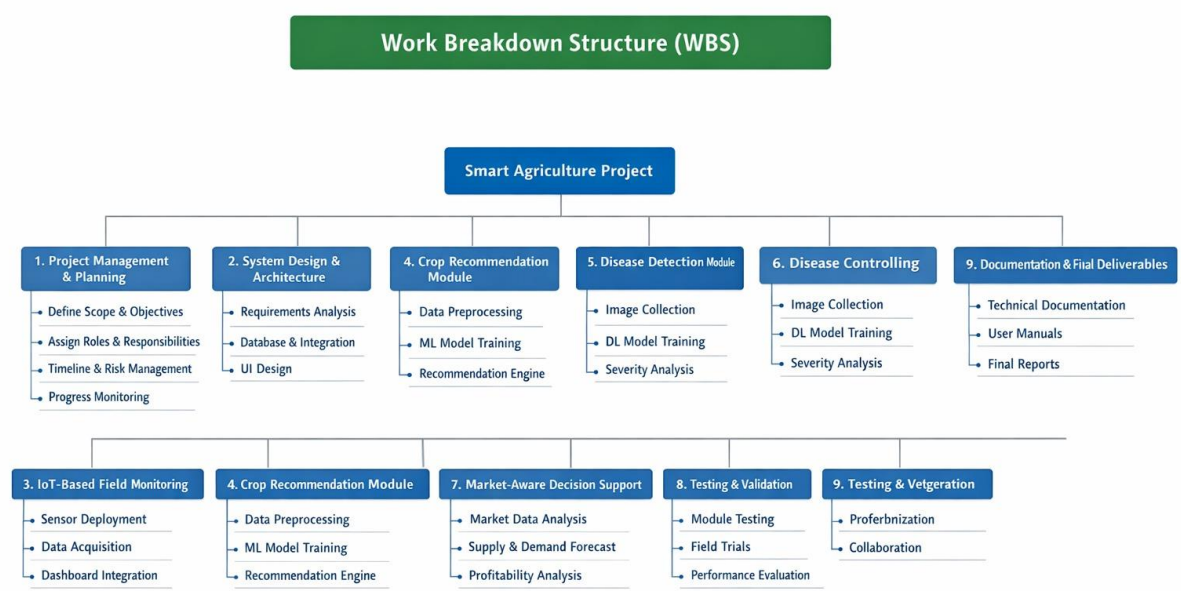
- The system shall allow **easy updating of machine learning models and market datasets** without affecting other system components.

7. BUDGET AND JUSTIFICATION

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application. Following is the budget justification for the whole research system.

Description	Cost
Electricity	5000.00
Internet	5000.00
Communication	2000.00
Transportation	2500.00
Paper Publishing	15000.00
Backend Hosting	12000.00
IoT Sensors	12000.00
Software Purchasing	8000.00
Total	52700.00

8. WORK BREAKDOWN STRUCTURE (WBS)



9. REFERENCES LIST

[1] Department of Agriculture - Sri Lanka, "Crop Recommendations and Fertilizer Guidelines," 2021.

[2] Department of Census and Statistics Sri Lanka, "Paddy Statistics - Maha 2023/2024 and Yala 2024," Ministry of Agriculture, Colombo, 2024. [Online]. Available: https://www.statistics.gov.lk/Agriculture/StaticalInformation/PaddyStatistics

[3] IRRI, "Standard Evaluation System (SES) for Rice," International Rice Research Institute, Los Banos, Philippines, 5th ed., 2013.

[4] S. Mandal et al., "Spectral characterization and severity assessment of rice blast disease using multispectral and hyperspectral indices," Frontiers in Plant Science, vol. 14, 2023, doi: 10.3389/fpls.2023.1072573.

[5] C. De Silva et al., "eKeth: A Machine Learning-Based Mobile Platform to Facilitate the Paddy Cultivation Process in Sri Lanka," in Proc. 6th Int. Conf. Information Technology Research (ICITR), Moratuwa, 10.1109/ICITR54349.2021.9657468. Sri Lanka, Dec. 2021, pp. 1-6, doi:

[6] G. Amarnath et al., "GeoGoviya: Smart Farming Platform for Climate-Smart Agriculture in Sri Lanka," International Water Management Institute (IWMI), Colombo, 2022. [Online]. Available: https://www.iwmi.org/projects/geogoviya/

[7] Department of Agriculture Sri Lanka, "Krushi Advisor: Agricultural Advisory Mobile Application," NAICC, 2023. [Online]. Available: https://doa.gov.lk/naicc-services-ict programs-mobile-applications/\

[8] Y. Wang, U. S. R. Dhamodharan, N. Sarwar, F. A. Almalki, and D. Mohan, "A hybrid approach for rice crop disease detection in agricultural IoT system," Discover Sustainability, vol. 5, art. no. 99, 2024, doi: 10.1007/s43621-024-00280-5.

[9] I. Abdurrahman and A. M. Miyim, "Machine Learning Model for Predicting Rice Crop Yield: A Case Study in Hadejia and Auyo, Nigeria," UMYU Scientifica, vol. 4, no. 1, pp. 239–249, 2025.

[10] W. Chen et al., "Deep learning-based identification and severity assessment of rice blast disease," Computers and Electronics in Agriculture, vol. 220, art. 108901, 2024.

[11] A. Gumma et al., "Satellite-based assessment of crop growth stages in paddy dominated Asian lowlands," International Journal of Remote Sensing, vol. 42, no. 5, pp. 1795 - 1816, 2021.

[12] D. G. Peng et al., "Modeling the impact of rice blast on yield under climate change scenarios in Asia," Global Food Security, vol. 28, art. 100493, 2021.

APPENDICES

Survey Instruments

- **Farmer Questionnaire**:
    - Current crop selection methods (experience-based, guideline-based).
    - Awareness and willingness to adopt IoT-based recommendations.
- **Agricultural Officer Interview Guide**:
    - Insights on regional soil/climate conditions.
    - Observations on farmer decision-making patterns.

Detailed Diagrams

- **System Architecture Diagram**:
    - IoT sensor layer (soil pH, NPK, moisture, temperature).
    - Climate data integration (weather forecasts, rainfall patterns).
    - ML-based recommendation engine.
    - Farmer-facing web dashboard.
- **Data Flow Diagram (DFD)**:
    - Input: IoT + climate
    - Processing: Preprocessing → Feature engineering → ML model.

- Output: Ranked crop recommendations with profitability scores.

Preliminary UI Mock-ups

- **Farmer Dashboard**:
  - Crop recommendation list with profitability index.
  - Soil condition summary (real-time sensor readings).
- **Recommendation Detail Page**:
  - Agronomic suitability score.
  - Market viability score.
  - Suggested cultivation timeline.

Risk Analysis

Risk	Likelihood	Impact	Mitigation
Inaccurate market forecasts	Medium	High	Use multiple data sources, update frequently
Sensor malfunction/data loss	High	Medium	Redundant sensors, error-checking algorithms
Farmer resistance to adoption	Medium	High	Training workshops, simplified UI
Data privacy concerns	Low	High	Secure storage, anonymization of farmer data
Climate anomalies (e.g., cyclones)	Medium	High	Integrate real-time weather alerts

Ethical Documents

- **Informed Consent Form** for farmers participating in surveys and field trials.
- **Data Privacy Policy** ensuring farmer data (soil, yield, market decisions) is anonymized and securely stored.
- **Fair Use Statement** clarifying that recommendations are advisory, not mandatory, and farmers retain autonomy in decision-making.
- **Bias Mitigation Plan** to ensure ML models do not favor specific regions or crop varieties unfairly.